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On the three-dimensional mechanical behavior of human breast tissue

Christian Goodbrake^a, David S. Li^{a,b}, Hossein Aghakhani^a, Alejandro Contreras^c, Gregory P. Reece^d, Mia K. Markey^{b,e}, Michael S. Sacks^{a,b,*}

^aJames T. Willerson Center for Cardiovascular Modeling and Simulation, Oden Institute for Computational Engineering and Sciences, Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712, USA

^bDepartment of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712, USA

^cDepartment of Pathology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

^dDepartment of Plastic Surgery, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

^eDepartment of Imaging Physics, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

Abstract

As the human breast undergoes complex, large-scale, fully three dimensional deformations in vivo, three-dimensional (3D) characterization of its mechanical behavior is fundamental to its diagnosis, treatment, and surgical modifications. Its anisotropic, heterogeneous fibrous structure results in complex behavior at both the tissue and organ levels. Mathematically modeling of this complex anisotropic behavior is thus critical to the proper simulation of the human breast. Yet, current breast tissue constitutive models do not account for these complexities, so that there is a pressing need for more detailed fully 3D analysis. To this end, we performed a full 3D kinematic mechanical evaluation of human fibroglandular and adipose breast tissues. We utilized our recently developed 3D kinematic numerical-experimental approach to acquire force-displacement data from both breast tissue subtypes. This was done by subjecting cuboidal test specimens, aligned to the anatomical axes, to both pure shear and simple compression loading paths. We then developed novel constitutive model that was able to simulate the unique anisotropic tension/compression behaviors observed. Constitutive model parameters were determined using a detailed finite element model of the experimental setup coupled to nonlinear optimization. We found that human breast tissues displayed complex anisotropic behavior, with strong, directionally dependent non-linearities. This was especially true for the fibroglandular tissue. The novel constitutive model was also able fully capture these behaviors, including states of combined tension and compression

*Corresponding Author: Michael S. Sacks (msacks@oden.utexas.edu).

Conflicts of interest

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(i.e. in pure shear). The results of this study suggest that human breast tissue is complex in its mechanical response, exhibiting varying levels of anisotropy. Future studies will be required to link the observed anisotropy to the physical structure of the tissue, as well as mapping this heterogeneity and anisotropy across individuals.

Keywords

breast; mechanical testing; finite element modeling; constitutive modeling

1. INTRODUCTION

The human body is composed of a variety of complex soft tissues whose mechanical properties are determined by their structure and composition Fung [2013]. In general, constitutive modeling can provide guidance for diagnosis of diseases and the design of tissue modification and its replacement [Krouskop et al., 1998]. Characterization and constitutive modeling of these tissues remains a challenging task both regards to theoretical formulation and in parameter identification. Typically, soft tissue studies are conducted with combinations of mechanical tests, morphological measurements, and computational modeling, all performed at the meso scale (at lengths of 3–10 mm).

One prominent area in which detailed knowledge of tissue-level behaviors remains limited is that of the human breast [Gefen and Dilmoney, 2007]. Breast cancer is a leading cause of death in women [Oeffinger et al., 2015], and the standard medical practice for detecting and evaluating breast cancer is screening and diagnostic imaging, predominantly in the form of mammography, and breast tomosynthesis. Detected tumors display different physical properties, namely they behave more stiffly than healthy tissue. This has brought forth a new imaging modality elastography [Krouskop et al., 1998], with the aim to identify regions of the tissue which exhibit pathological characteristics. Treatment of cancer may require complete total mastectomy, or may involve surgery to remove of the tumor and surrounding tissue (segmental mastectomy), followed by breast reconstruction [Caplan, 2014, Omid et al., 2014] and radiation therapy. Regardless of the technique used post-removal, any implant and/or reconstructed patient tissues must be able to support the physiological mechanical forces acting on the breast. Otherwise, mechanical failure of the implant or necrosis of the tissue may occur [Gefen and Dilmoney, 2007]. Detailed understanding about the biomechanical behaviors of breast tissue, especially its suspensory ligament system, can assist in post-surgical breast reconstruction and implant design [Pathmanathan et al., 2004].

Human breast tissue is known to have a complex, heterogeneous composition that varies both among individuals and over time [Gefen and Dilmoney, 2007]. The effect of this composition on the tissue's mechanical behavior however, is not thoroughly investigated. Understanding the three dimensional (3D) biomechanical properties of breast tissue can benefit breast cancer treatment in the context of surgical guidance, as well as aid in diagnostic tools that require accurate simulation of patient specific organ behavior, and has the potential to drive implant design in the future. The better guidance of the medical treatment of the breast relies upon an accurate, patient-specific geometric and mechanical

model to guide the reconstruction of the breast following segmental mastectomy, or predict how the surrounding tissue will respond to the procedure.

To date, the reported mechanical properties of the breast are generally based on simplified uniaxial compression tests and are modeled using isotropic constitutive relations [Dempsey et al., 2021, Gefen and Dilmoney, 2007, Omid et al., 2014]. These tests often do not consider the anatomical orientation of samples, assume isotropy, or as in the case of Dempsey et al. [2021] cannot distinguish between isotropic and anisotropic samples. However, the anatomical structure of the breast could drive anisotropy within the tissue, especially in the suspensory ligaments and fibroglandular regions [Gefen and Dilmoney, 2007]. Despite this, there is limited work in characterization of this tissue, both in terms of available mechanical data as well as extant material models. In particular, existing tissue evaluation relies on uniaxial tensile testing; the imposition of different deformation modes in full 3D is utterly lacking. Crucially, the multiaxial compression behavior of the tissue has not been well studied, despite the central role this deformation mode plays in mammography, for instance.

It is also still unclear whether human breast tissue is mechanically anisotropic at the meso (~1-cm) scale, and if so, to what degree and in what anatomical orientations. Microstructural analysis of the breast suggests the presence of constituents like suspensory ligaments and glandular tissue that may influence an anisotropic response [Gefen and Dilmoney, 2007], yet this remains to be evaluated experimentally, nor is any existing anisotropy accounted for current models of breast tissue [Samani and Plewes, 2004]. Moreover, authors studying other tissues in compression have found success employing bimodular models to describe asymmetries in tension and compression [Klisch, 2007]. It is unknown whether such approaches are required to model breast tissue. Even in the case of isotropic nonlinear materials, there is no theoretical limit to the number of parameters required to fully characterize their behavior. This discrepancy is only magnified for anisotropy, so we must perform multiaxial measurements to capture the full behavior of human breast tissue.

Therefore, in the present study, we sought to characterize human breast tissue's full 3D mechanical behavior. We utilized a recently developed 3D multi-axial integrated numerical-experimental approach [Avazmohammadi et al., 2017] to fully simulate human breast tissue mechanical responses. Notable features of this approach included the ability to apply all displacement paths to a single specimen, use of a high-fidelity FE-based simulation of the experimental configuration, and enhanced determinability and predictive capability of the estimated parameters using an inverse modeling technique. This model then allows us to gain a more detailed and quantitative insight into the 3D mechanical behavior of the human breast.

2. METHODS

2.1. Overall approach

Our goal herein is the elucidation of three dimensional (3D) mechanical properties of human breast tissue and development of a novel constitutive model for its unique behaviors. We utilized our full 3D numerical-experimental approach [Avazmohammadi et

al., 2018] to provide the necessary experimental data and parameter estimation pipeline. We accomplished this by first acquiring human cuboidal breast tissue samples from a cohort of breast cancer patients, making sure to take samples from healthy portions of the breast away from tumor sites. These samples were then subjected to several deformation protocols using a triaxial testing system. Based on initial observations of the resulting data, we formulated a constitutive model, and utilized finite element analysis optimization to determine the optimal values of the model parameters. We also verified the predictive capability, based on a subset of the measured deformation protocols. This fitted model was then analysed to determine its degree of anisotropy which we can then use to infer the inherent anisotropy of the tissue itself.

2.2. Tissue sourcing and anatomical selection

Normal human breast tissue samples were collected from a cohort of breast cancer patients undergoing breast surgery, in compliance with the institutional review board at The University of Texas MD Anderson Cancer Center (PA16-0364). Anatomically oriented 2×2×2-cm samples were obtained from multiple healthy regions in the breast in order to bracket spatial variations in mechanical properties and classified into “Adipose” and “Fibroglandular” groups by our pathologist depending on their relation to physical breast anatomy (n=3 for each group). Specimens were first inert dye marked to identify the anatomic axes (Figure 1-A). Before testing, each specimen was further trimmed to 1×1×1-cm cuboids using microtome blades, with the edges of the specimens aligned to the medial-lateral (Med-Lat), superior-inferior (Sup-Inf), and anterior-posterior (Ant-Pos) directions of the breast (Figure 1-B).

2.3. Experimental setup

To measure the full 3D mechanical response of the tissue, we used a previously developed triaxial testing technique [Avazmohammadi et al., 2018]. Briefly, specimens were mounted by attaching nine-pin arrays to each face of the cuboid with cyanoacrylate gel-based adhesive (Figure 1-C)). Load cells within the attachment systems of the device were used to record the 3D force response of the tissue in real time as deformations were applied, controlled with a custom LabVIEW virtual instrument (National Instruments, TX, USA). Due to the pin attachments of our samples, the deformation of each sample deviated from the homogeneous deformation we sought to impose, since, while the displacement boundary conditions on the pins are consistent with these homogeneous deformations, the zero traction boundary conditions away from the pins allow each specimen side to deform relatively freely. Therefore, these deformations must be simulated numerically rather than assessed analytically. This approach allows for a simple specimen attachment setup but more importantly complete accounting for any fibrous specimen heterogeneity Li et al. [2020b,c]. For simplicity, we report the homogeneous deformation gradient consistent with the imposed pin displacements, with these being identified with the X or 1, 2 or Y, and 3 or Z directions respectively for orientation of the specimen within the mechanical testing device (Figure 1-C)). All testing was performed with the specimens immersed in phosphate buffered saline maintained at 37°C.

2.4. Deformation protocols

Bearing in mind the use of an inverse modeling approach required and the relatively unknown properties of the human breast tissue, deformation protocols were chosen to simultaneously explore the multi-axial behavior of the tissue and produce the requisite data for robust parameter estimation (Table 1). Specifically, the tissue's mechanical response was assessed in both two-axis tension-compression deformations in which one axis is extended while another axis is simultaneously compressed such that the unconfined third axis remains unstretched and single-axis (unconfined) compression (Figure 2). This is known as a 'pure shear' mode. Additionally, these loading paths were imposed on varying combinations of specimen axes to differentiate the tissue's response with respect to anatomical orientation. We further note that use of simultaneous states of tension and compression allowed determination of bimodular material behavior. In addition, we conducted simple compression tests along each anatomical axis (Figure 2). This comprehensive multiaxial evaluation was necessary, since previous studies focused primarily on the tensile behavior of human breast tissue, despite the central role of compression in mammography and elastography techniques. In practice, each protocol was repeated over 10 cycles to precondition the tissue.

2.5. Data post-processing

We note that while some variability in the experimental data occurred, we observed a high degree of consistency in the same anatomical location sub-group, with a standard error of $\leq 20\%$ at peak deformation. As our main focus was on determining an appropriate form and reasonable estimate of the material parameters, we averaged the data from each individual protocol for each sub-group to create tissue group averaged response. This approach had the added benefit of reducing the effect of measurement noise a priori, and has been successfully used by our group in previous soft tissue studies to develop accurate group responses Sacks and Chuong [1998].

We note too that due to Central Limit Theorem and assuming that the model parameters are independent random variables, uncertainty in the averaged data can be well represented in form of Gaussian distributions, even if the uncertainty in the original data cannot be. The functional independence of the different parameters appearing in our model is established in Appendix C. Solving the inverse problem using the average of the experimental results for each test protocol thus leads to a maximum a posterior estimation of the final distribution. The study of the input uncertainties on the final distribution is very important but is out of scope of the current work, as obtaining concrete estimates of the distribution of physical properties across the population would require a greater number of specimens. This averaged data $\hat{\mathbf{S}}$

was then used as the basis for all parameter fitting and validation.

2.6. Constitutive model

Because we sought to model any observed inherent anisotropy that may be present in human breast tissues, we decided to choose an appropriately general model form that would allow simulate a range of fit isotropic to fully anisotropic behaviors, while also facilitating

quantification of the degree of observed anisotropy. As is commonly done in the modeling of soft tissues, we assumed that the size of the specimen ensured the mechanical and structural properties were approximately homogeneous at the length scale of study. The deformations were represented by a map

$$\mathbf{x} = \varphi(\mathbf{X})$$

, where

$\mathbf{x}$

parameterizes the loaded configuration, and

$\mathbf{X}$

parameterizes the unloaded configuration. The breast was modeled as a (pseudo)-hyperelastic material Fung [2013] whose deformation can be locally characterized by the deformation gradient tensor

$$\mathbf{F} = \text{Grad } \mathbf{x}$$

. We represented the strain energy in the tissue

$$\psi(\mathbf{E})$$

as a function of the Green-Lagrange strain tensor

$$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$$

, which induces a second Piola stress tensor

$$\mathbf{S} = \frac{\partial \psi}{\partial \mathbf{E}}$$

. is common with biological materials, we assume incompressibility because of the high water content present in the tissue, requiring a Lagrange multiplier that augments the stress

$$\mathbf{S} = \frac{\partial \psi}{\partial \mathbf{E}} - p \frac{\partial (J - 1)}{\partial \mathbf{E}}, \quad (1)$$

where

$$J - 1 = \text{Det } \mathbf{F} - 1 = \sqrt{\text{Det } (2\mathbf{E} + \mathbf{I})} - 1 = 0$$

is the incompressibility constraint. For computational purposes, we replace the Lagrange multiplier term with an energy penalty term of the form

$$K (J - 1)^2$$

. Specifically, we adopt a modified anisotropic nonlinear neo-Hookean strain energy density of the form

$$\psi(\mathbf{E}) = \mu \text{Tr}(\mathbf{E}) + \mathbf{E} \cdot \mathcal{M}[\mathbf{E}] + a \exp\left(\frac{b}{2} E_{33}^2\right) + K (J - 1)^2 \quad (2)$$

where

$$\text{Tr}(\mathbf{E}) = \frac{1}{2} (I_1 - 3)$$

, and

$$\boldsymbol{\alpha} = \{\mu, \mathcal{M}, K, a, b\}$$

are the material model parameters,

μ

being a neo-Hookean like shear modulus in the small strain limit,

$\mathcal{M}$

being a fourth order tensor of material parameters,

K

being a bulk modulus, and

a

and

b

determining the stronger nonlinearity with respect to straining in the Ant-Pos

(

x_3

) direction. This form was inspired by early Fung models, which were the sum of a quadratic form and the exponential of another quadratic form [Fung, 2013]. We add a linear neo-hookean term and an incompressibility constraint to model possible tension-compression asymmetry, and restrict the two quadratic forms to reflect our observed anisotropy. To simulate near incompressibility, we let

$$\frac{K}{\mu} \gg 1$$

so that the volumetric penalty is the dominant contribution to the energy, even though the bulk modulus is still a parameter that we are fitting. This form was chosen based on a preliminary examination of the experimental fibroglandular data, which revealed a strong nonlinearity in the Ant-Pos direction (here identified with the

x_3

coordinate) that was not present in the other directions, together with otherwise mild anisotropy. For the adipose tissue, the exponential term did not improve fits, and accordingly, we take

$$a = 0$$

for adipose tissue, which also eliminates

b

from the model.

Additionally, in choosing the form of

$\mathcal{M}$

we utilized the hypothesis that the breast tissue is orthotropic with respect to the anatomical axes, as we expect that any anisotropy present in human breast tissue most strongly present with respect to these axes. Therefore, by restricting the model's form to orthotropy with respect to these axes, we captured the dominant portion of anatomical anisotropy. A secondary advantage of this is that we reduce the number of material parameters in our model, which both prevents overfitting and reduces the computational cost of optimizing parameters.

Explicitly in Voigt notation, the Green's strain tensor can be written as

$$\mathbf{E} = [E_{11} \ E_{22} \ E_{33} \ E_{23} \ E_{13} \ E_{12}]^T, \quad (3)$$

after which, the tensor of material parameters can be written as

$$\mathcal{M} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & 0 & 0 & 0 \\ M_{12} & M_{22} & M_{23} & 0 & 0 & 0 \\ M_{13} & M_{23} & M_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{66} \end{bmatrix} \quad (4)$$

This term, being the quadratic coefficient of the strain energy, can be directly identified with the usual tensor of elastic moduli in linear elasticity, and hence can be decomposed into an isotropic component, and a purely anisotropic component. This allows us to quantify the degree of low order anisotropy by considering the fraction of the norm of the tensor of elastic moduli contributed by the anisotropic component. We note additionally that when making this identification, the term

M_{33}

must be augmented with the addition of

ab

and a volumetric term involving

K

to account for the low order contribution of the exponential term and incompressibility penalty, since the expansion of the energy to quadratic order is equal to

$$\psi = \mu \operatorname{Tr}(\mathbf{E}) + \mathbf{E} \cdot \mathcal{M}[\mathbf{E}] + abE_{33}^2 + K(E_{11}^2 + E_{22}^2 + E_{33}^2 + 2(E_{11}E_{22} + E_{22}E_{33} + E_{33}E_{11})) + O(\mathbf{E}^3). \quad (5)$$

Finally we note that the presence of a linear term in the strain energy density would suggest the presence of residual stress in the reference configuration, since this term directly corresponds to stress in the unloaded configuration. The isotropy of the linear term results in a residual stress that is a pure pressure. This means that the residual stress that would be predicted is counteracted by the incompressibility constraint, and hence this linear term does not create residual stress in the reference configuration. This term does however, create a tension-compression asymmetry as demonstrated in Appendix B, since it is an odd function of the Lagrange strain, eliminating the need to employ a bimodular theory as was done in Klisch [2007].

2.7. The Forward Problem

Due to the nature of the inverse problem that necessitates many forward solves plus the complex nonlinearity in the model finding an efficient solver is crucial. Toward this aim, a Newton-Krylov solver preconditioned with Algebraic Multigrid (AMG) was used with GMRES kernel update every 30 iterations. For smaller deformations a relaxation factor equal to 1 worked well, but for larger deformations reducing the relaxation factor to 0.5 was necessary to handle the increasing nonlinearities. Our experience showed non-preconditioned Krylov solvers were incapable of solving the linear problem, and direct solvers were too expensive to compute as expected. Moreover, an advantage of using

iterative solvers for optimization problems is the use of the strong Wolfe condition which allows adaptive updating of the convergence criteria while the optimizer gets closer to the target. In light of the boundary conditions resulting from the nine-pin attachments in mechanical testing, we employed a finite element (FE) representation of the tissue specimen mounted to the triaxial testing device, featuring a volume mesh of tetrahedral elements with surface node subsets on each cuboid face to replicate the nine-pin attachments. Details of this approach are given in Avazmohammadi et al. [2018].

2.8. Inverse modeling, parameter estimation, and validation

The optimal material parameters

α

for both constitutive model forms (i.e. including or excluding the exponential term) were computed with an inverse modeling framework based on our previous approaches [Avazmohammadi et al., 2018, Li et al., 2020a]. A subset of the tri-axial deformations containing the pure shear protocols and one simple compression protocol was simulated in parallel forward simulations, with each loading path in the set prescribed as Dirichlet boundary conditions to the attachments. The modeled stress $S(\alpha)$

was computed for a given set of material parameters and compared to the experimentally measured data

$\hat{S}$

with the cost function (Eq. 6)

$$\Phi(\alpha) = \sum_{m=1}^{\text{Modes}} \sum_{n=1}^{\text{Faces}} \sum_{q=1}^{\text{Data}} \|\hat{S} - S(\alpha)\|_{m,n,q}, \quad (6)$$

where we sum over the indices

m, n

and

q

corresponding to the deformation modes, relevant cuboid faces, and experimental data points, and

$\|\cdot\|$

is the

L^2

norm. The optimal parameters

α^*

were estimated by minimizing

Φ

(Eq. 7):

$$\alpha^* := \underset{\alpha}{\operatorname{argmin}} \Phi(\alpha). \quad (7)$$

with the reflective trust region method. Gradients were computed by the central finite difference scheme fed with complex numbers. By using complex numbers both the initial value of the governing PDE and its perturbed value can be computed at a point with a single evaluation which nearly (though not exactly, due to the additional overhead associated with using complex instead of floating point numbers) halves the computational cost. The trust-region size was updated iteratively as needed, especially when the optimizer was far from the optimal parameters. A simplified single element analytical solution was used calculate an initial guess for the optimal parameters, a critical step in the convergence of any Newton-based method. As an extension to our previous approaches, we converted the entire finite element simulations to the open source computing platform FEniCS (fenicsproject.org). We used the Stampede system at the Texas Advanced Computing Center. Once the set of optimal parameters α^*

is determined, the remaining triaxial deformations are simulated, and the predicted results are compared with the experimental data, to demonstrate the predictive capacity of our model.

3. RESULTS

3.1. General observations

The stretch stress-stretch responses from all protocols applied to adipose and fibroglandular breast tissue specimens exhibited mild nonlinearity across all deformation modes. Beyond 20-30% deformation, the initially soft and mostly linear curves of protocols 1 – 6 showed a highly nonlinear increase, consistent with other work on adipose tissue mechanical response. Examining the data from multiple cycles showed that preconditioning effects were most prevalent in the first cycle, whereas subsequent cycles had a more reproducible response with less hysteresis.

Although the response of the adipose tissue was only mildly anisotropic, mechanical testing results revealed stronger anisotropy in the fibroglandular specimens, as can be observed from the disparate high-strain stresses obtained protocol 6 in comparison to protocols 4 – 5. In particular, deformations along the Ant-Pos direction showed greater maximum stress compared to other directions. The fitted parameters for both models are presented in Table 2. Additionally, these parameters were able to fit both compressive and tension behavior for both tissues, indicating that a bimodular model was unnecessary.

3.2. Adipose tissue

The mildly anisotropic model was able to achieve a reasonable fit to the mechanical data of adipose tissue specimens (Fig 3a). The fitted model predicts the stress results of the deformation modes not included in the fitting data reasonably well (Fig 3b).

3.3. Fibroglandular tissue

The overall mechanical response of fibroglandular tissue differed from that of the adipose tissue specimens, achieving greater maximum stresses in tension-compression modes but generally lower maximum stresses in compression. There was also a more pronounced

anisotropic mechanical profile across protocols, compared to adipose tissue. The addition of the anisotropic exponential term allowed our model to fit the data well (Fig 4a). Additionally, this fitted model was able to predict the stress in deformation modes outside of the fitting data (Fig 4b).

3.4. Compressive behavior of breast tissue

The form of the fitted model indicates that fibroglandular tissue exhibits a strong stress-strain nonlinearity with respect to simple compression in the Ant-Pos direction in comparison to either the Med-Lat or Sup-Inf directions (Fig 5). This initially appears at odds with the common convention regarding fibrous tissues, which is that fibers are incapable of bearing compressive loads. However, the incompressibility constraint allows us to recast this term as a function of the positive strains in the Med-Lat and Sup-Inf direction. Hence if this nonlinearity is due to the fibrous content, the form of the data suggests that these fibers lie in the plane spanned by the Med-Lat and Sup-Inf directions.

3.5. Anisotropy

We seek to quantify if there are differences in the anisotropy present in adipose compared to fibroglandular tissues. We do this by examining the stiffness matrix present in both models, and determine the relative degree of anisotropy present in these matrices. For comparison, the most general isotropic quadratic function of the Green Lagrange strain has a coefficient matrix that takes the form

$$\mathcal{M}_{iso} = \begin{bmatrix} K + \frac{4}{3}\mu & K - \frac{2}{3}\mu & K - \frac{2}{3}\mu & 0 & 0 & 0 \\ K - \frac{2}{3}\mu & K + \frac{4}{3}\mu & K - \frac{2}{3}\mu & 0 & 0 & 0 \\ K - \frac{2}{3}\mu & K - \frac{2}{3}\mu & K + \frac{4}{3}\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix}. \quad (8)$$

Stiffness matrices themselves span a vector space, with the matrices corresponding to isotropic materials spanning a two-dimensional subspace of this vector space. We want to decompose the total space into a purely isotropic subspace, and its orthogonal complement, which we refer to as the anisotropic subspace. We can then project our optimized stiffness matrices onto these two subspaces and compare the relative magnitude of their anisotropic components. The definition of the orthogonal complement and the projection onto these subspaces was accomplished using the natural inner product induced by the referential metric tensor, which differs from the Frobenius norm on 6x6 matrices that one might naively employ.

With this norm, we take the quadratic component of our fitted strain energy, and compute its projection onto the two-dimensional isotropic subspace, and the anisotropic subspace in a way that agrees with the intrinsic geometry of our problem. We then express our model's

quadratic coefficient tensor as the sum of an isotropic term and an anisotropic term. This allowed us determine the degree of our model's anisotropy at quadratic order by computing the magnitude of the anisotropic component and comparing it to the total norm of the coefficient tensor. Performing this analysis, we determined that our fitted fibroglandular model's relative anisotropy is 2.04 times the relative anisotropy of the adipose model, providing a quantitative measure of how much more anisotropic fibroglandular tissue is than adipose tissue.

Additionally, we note that the vast majority of each model's coefficient tensor (over 99%) lies in the subspace parameterized by the bulk modulus, indicating that the assumption of incompressibility for both of these tissues is well justified. Removing this component, we find that the overwhelming majority of both models' remaining strain energy is anisotropic (99.9% for adipose tissue, and 95.8% for fibroglandular). One might expect that these deviatoric anisotropies would mimic the pattern we see in the total anisotropies, in that we would expect that the fibroglandular tissue's deviatoric anisotropy would be higher than that of the adipose tissue, however here we observe a reversal. This indicates that while the isotropic component of both tissues is primarily due to incompressibility, the fibroglandular tissue has a comparatively larger isotropic stiffness in shear, relative to the adipose tissue. Therefore, when the volumetric component is totally removed, the fibroglandular tissue's energy possesses a comparatively larger remaining isotropic component, and hence a smaller deviatoric anisotropy. Furthermore, we found that the fibroglandular tissue displayed stronger anisotropy at higher strains than the adipose tissue, which we modeled by the imposition of the exponential term. While did not rigorously quantified the anisotropy of this term as we have with the rest of the parameters in the model since it does not lie in a clearly defined finite dimensional vector space, we note that it is transversely isotropic with the Ant-Pos direction as its distinguished direction. Overall, these results indicate that beyond the incompressibility constraint, which is isotropic, these tissues display complicated anisotropic behavior that cannot be captured by an isotropic model.

4. DISCUSSION

4.1. General findings

In the course of this study, we sought to examine and quantify the mechanical behavior of human breast tissue, and in particular, we sought to assess and quantify the anisotropy present in this behavior. Additionally, we specifically wanted to capture breast tissue's behavior in compression along the anatomical axes, since most previous studies only studied breast tissue's behavior in tension. We did this by taking samples classified as either "adipose" or "fibroglandular," imposed multiaxial deformations upon these, and then fit an anisotropic model to each of these collected data sets. Based on these fitted models, we quantified the relative incompressibility and anisotropy of these tissue types, and found that these tissues can be modeled as incompressible, and that their anisotropy could not be neglected. Not only did we find that both tissue types were anisotropic, we found that fibroglandular tissue displayed substantially stronger anisotropy, necessitating an additional exponential term. Despite this feature, a bimodular model was unnecessary as both compressive and tensile components of stress were well predicted by both models. These

results suggest that properly identifying the orientation of strong anisotropy in breast tissue arising from its microstructure significantly improves robustness of its characterization.

We also found qualitative differences in the mechanical behavior of our fitted models. Under the imposition of incompressibility, our fitted adipose model's energy has a minimum at zero strain, while our fitted fibroglandular model's energy has a saddle point at zero strain, with local minima nearby. Outside of a small region around zero strain, both of our models' fitted energies are convex. This allows our fibroglandular model to behave incredibly softly at low strains, matching our experimental data, while providing stability at larger strains. While this lack of convexity at zero strain may initially seem concerning, the difference between the energy at zero strain and the minimum energy is very slight, which yields a model that is capable of capturing the softness of fibroglandular tissue at low strain. Furthermore, because the region of fibroglandular nonconvexity is small, we retain the advantages of a convex model at finite strains; our adipose model is convex so we don't encounter a similar issue with adipose tissue. This analysis is presented in Appendix A.

4.2. Links to breast tissue structure and organ level function

Our results suggest that human breast tissue is complex in its mechanical response, exhibiting both anisotropy to various degrees and heterogeneity. Additionally, linking this anisotropy to the physical structure of the tissue is necessary; validation with histological analysis of the specimens remains to be performed. Furthermore, the mechanical and structural differences between adipose and fibroglandular regions have yet to be more thoroughly assessed, and more rigorous comparison of the constitutive models have yet to be performed with metrics like the Akaike information criterion. This suggests that further studies more thoroughly mapping this heterogeneity and anisotropy, as well as quantifying the variance in material properties and structure across individuals are necessary. Additionally, a more detailed characterization of the in vivo mechanical environment of breast tissue can serve to guide these studies more efficiently. By performing these studies and having a thorough understanding of the complex mechanical behavior of human breast tissue, both at the general population level and the patient-specific level, we can accurately predict healthy tissue's behavior during deformation-based diagnostic techniques, and hence identify pathological behavior. This information will also improve confidence in the accuracy of simulation data, such as internal stresses and strains, when direct measurement is impossible or invasive. Knowledge of these internal stresses and strains experienced by the breast can be used to guide the development and systematic improvement of tissue reconstruction techniques, synthetic implants, and surgical interventions.

4.3. Summary

In this study, we developed the first specialized 3D model of the mechanical behavior of human breast tissue using comprehensive 3D numerical-experimental approach. This novel model of adipose and fibroglandular specimens revealed region-dependent differences in tissue mechanical properties, which supports the need for incorporating heterogeneity in whole-breast simulations. Additionally, the form of this model allowed us to quantify the degree of anisotropy in both tissues, allowing us to precisely compare the qualitative differences in the fitted models. Despite this, we did not find it necessary to employ a

bimodular model to capture the behavior of either adipose or fibroglandular tissue. Our fitted parameters for both models suggests that both adipose and fibroglandular tissues are effectively incompressible, and so incompressible models are well justified. The fitting of anisotropic models also indicates that increasing fibrous content in the breast tissue could indeed drive mechanical anisotropy at higher strains; however, for low fibrous content and at low tissue strains, the anisotropy of the tissue is well described as a low order phenomenon. We found that this anisotropy in all cases constitutes the majority of human breast tissue's isochoric response, hence perhaps the primary finding of our work is the dominance of breast anisotropy in volume-preserving motion. The findings presented in this work serve as a critical first step toward physiologically accurate 3D modeling of the breast and highlight important considerations for the constitutive modeling of soft tissues.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A. Convexity Analysis

We examined the convexity of our fitted models by examining the structure of their extrema. We do this under the assumption of incompressibility by the method of Lagrange multipliers. In effect, we seek to solve the system

$$\text{Grad}(\psi - \lambda\phi) = 0, \quad (\text{A.1})$$

where

ψ

is our strain energy,

λ

is a Lagrange multiplier,

ϕ

is the incompressibility constraint, and Grad is taken over the variable set

{

$E_{11}, E_{22}, E_{33}, E_{12}, E_{13}, E_{23}, \lambda$

} . Solutions to this system are stationary points of the energy, constrained to the

hypersurface of isochoric strains. Once we know these extremal points, we can examine the structure of the energy near these points to determine if they are local minima, maxima, or saddle points.

Appendix A.1. Adipose

Examining this system for the fitted adipose parameters, we find that there is a unique solution lying at the point of zero strain. It is not difficult to see that this point is a minimum; the incompressibility constraint can be solved for

E_{11}

for instance, and the energy constrained to isochoric motions by substituting this solution into the expression for the energy. Doing this, and taking the Hessian of this constrained energy density at the point of zero strain yields the Hessian matrix

$$\begin{bmatrix} 261.6 & -39.2 & 0 & 0 & 0 \\ -39.2 & 255.6 & 0 & 0 & 0 \\ 0 & 0 & 473.6 & 0 & 0 \\ 0 & 0 & 0 & 487.6 & 0 \\ 0 & 0 & 0 & 0 & 743.6 \end{bmatrix}, \quad (\text{A.2})$$

which is positive definite, having a minimum eigenvalue 219.285. Therefore, the zero strain state in the incompressible model is a global minimum.

Appendix A.2. Fibroglandular

We note that the unconstrained fibroglandular energy density can be written as

$$\psi = f(E_{11}, E_{22}, E_{33}) + 158E_{23}^2 + 242E_{13}^2 + 116E_{12}^2, \quad (\text{A.3})$$

which is clearly convex in the shear terms, since for fixed

E_{11}

,

E_{22}

,

E_{33}

, this function takes its minimum value when these shear components vanish. Hence, we simply have to examine the restriction of the energy when these shear terms vanish, i.e. the function

f

. Applying the incompressibility constraint allows us to represent

f

as a function of two independent strain components, say

E_{11}

and

E_{22}

. Plotting this constrained energy as a surface reveals two local minima, and a saddle point, the saddle point occurring at

$\mathbf{E} = \mathbf{0}$

with the constrained energy taking the value

$\psi = 1.59$

. The global minimum occurs at

$$\mathbf{E} = \text{diag}(0.460, -0.193, -0.075)$$

, with the energy taking a value of

$$\psi = -4.176$$

, and the other local minimum occurs at

$$\mathbf{E} = \text{diag}(-0.204, 0.175, 0.126)$$

with a value of

$$\psi = -0.665$$

. While this technically means that the fitted model is unstable at zero strain, the energy at zero strain is sufficiently near the global minimal energy. As a comparison, the model's energy at the equibiaxial strain of state

$$\mathbf{E} = \text{diag}(0.2, 0.2, -0.245)$$

is

$$\psi = 45.069$$

, meaning that putting the tissue in this strain state from the zero strain state requires over 7.5 times the energy released by taking the tissue from zero strain to its global minimum. This lack of convexity causes the tissue to behave very softly in regions near there local extrema, while behaving stiffly at strains far from these extrema, replicating the observed softness at low strains and stiffness at higher strains.

Appendix A.3. Fibroglandular Correction

The lack of convexity in the fibroglandular model can be removed by adding the term

$$\psi_{add} = \begin{bmatrix} E_{11} & E_{22} & E_{33} \end{bmatrix} \begin{bmatrix} 32.85 & -21.93 & -12.75 \\ -21.93 & 14.64 & 8.51 \\ -12.75 & 8.51 & 4.945 \end{bmatrix} \begin{bmatrix} E_{11} \\ E_{22} \\ E_{33} \end{bmatrix}, \quad (\text{A.4})$$

to the total energy. This amounts to a correction on the fitted parameters

$$M_{11}$$

,

$$M_{22}$$

,

$$M_{33}$$

,

$$M_{12}$$

,

$$M_{13}$$

, and

$$M_{23}$$

, such that the final incompressible model has a global minimum at zero strain. This correction was found by expressing the matrix

$\mathcal{M}$

in terms of its eigenbasis, and subsequently increasing the smallest eigenvalue until the three critical points merged into one. This modified matrix was then expressed back in terms of

the original strain components, from which the correction above was computed. It is important to note that this does not make

$\mathcal{M}$

positive definite, but it does make the restriction of the incompressible energy convex at the zero strain point. This correction term can be increased to completely remove the convexity from

$\mathcal{M}$

, which will guarantee a convex energy for all strains, though in practice, this is likely more than sufficient.

Appendix B. Tension-Compression Asymmetry

Here we demonstrate how a Neo-Hookean term coupled with incompressibility generates tension-compression asymmetry in pure shear.

For simplicity, consider an incompressible Neo-Hookean material in pure shear in the 1 – 2 plane. The energy is then

$$\psi = \frac{\mu}{2} \left(C_{11} + \frac{1}{C_{11}} - 2 \right). \quad (\text{B.1})$$

We consider a displacement

α

, with

$$C_{11} = 1 + \alpha$$

, and the energy becomes

$$\psi = \frac{\mu}{2} \left(\alpha + \frac{1}{1 + \alpha} - 1 \right) = \frac{\mu}{2} \left(\frac{\alpha^2}{1 + \alpha} \right). \quad (\text{B.2})$$

For positive

μ

, this has a minimum at

$$\alpha = 0$$

, hence our material is stable at zero strain, however, it is clear to see that

$$\psi(\alpha) = \frac{\mu}{2} \left(\frac{\alpha^2}{1 + \alpha} \right) \neq \psi(-\alpha) = \frac{\mu}{2} \left(\frac{\alpha^2}{1 - \alpha} \right). \quad (\text{B.3})$$

This asymmetry is

$$\psi(\alpha) - \psi(-\alpha) = \frac{\mu\alpha^2}{2} \left(\frac{1}{1 + \alpha} - \frac{1}{1 - \alpha} \right) = \frac{\mu\alpha^3}{\alpha^2 - 1}, \quad (\text{B.4})$$

therefore it is clear that the Neo-Hookean term under the constraint of incompressibility generates the observed tension-compression asymmetry.

Appendix C. Robustness Analysis

Due to the relatively large number of parameters appearing in our model, there may be concerns about parameter identifiability. We seek to establish the functional independence of these parameters and provide an a posteriori justification of their use. Recall that our proposed strain energy density takes the form

$$\psi = \mu \operatorname{Tr}(\mathbf{E}) + \mathbf{E} \cdot \mathcal{M}[\mathbf{E}] + a \exp\left(\frac{b}{2} E_{33}^2\right) + K(J-1)^2.$$

Ordering the material parameters as

$$\alpha = \{\mu, M_{11}, M_{22}, M_{33}, M_{12}, M_{13}, M_{23}, M_{44}, M_{55}, M_{66}, a, b, K\}$$

, the corresponding sensitivity functions take the form

$$\left[\operatorname{Tr}(\mathbf{E}), E_{11}^2, E_{22}^2, E_{33}^2, 2E_{11}E_{22}, 2E_{11}E_{33}, 2E_{22}E_{33}, E_{23}^2, E_{13}^2, E_{12}^2, \exp\left(\frac{b}{2} E_{33}^2\right), \frac{a}{2} \exp\left(\frac{b}{2} E_{33}^2\right) E_{33}^2, (J-1)^2 \right]$$

Note that these functions are independent of the parameters except for

a

and

b

, and the only sensitivities depending on

a

and

b

are the sensitivities with respect to

a

and

b

. Therefore, we can evaluate these functions at the fitted values for the fibroglandular tissues, and simply remove the two sensitivities corresponding to

a

and

b

when we want to analyze the adipose tissue.

We seek to quantify the independence of these functions over the strain range considered in this study. To do this, we consider the region of strain space containing our experimental data, namely we bound the normal components of the strain to be between -0.2 and 0.345 , corresponding to the normal displacements -2.25 mm and 3 mm respectively. Likewise, we bound the shear components to be between ± 0.265 corresponding to the maximum shear strain encountered in our study. We further restrict this region of strain space to those strains satisfying

$$J > 0$$

so that only physical strains are considered. We denote this domain

$\mathcal{D}$

. This domain then defines a Hilbert space, where the inner product is taken to be

$$\langle f, g \rangle = \int_{\mathcal{D}} f \bar{g} d\mathbf{E},$$

where the overbar denotes complex conjugation, though all functions we will consider are real-valued. Like any other inner product, this allows us to compute the magnitude of functions on this space as

$$\|f\|^2 = \langle f, f \rangle,$$

and allows the computation of angles between functions as

$$\cos \theta_{f,g} = \frac{\langle f, g \rangle}{\|f\| \|g\|}.$$

We do this for each pair of functions

f_i

,

f_j

, and obtain a matrix of cosines

$\mathbf{A}$

where

$$A_{ij} = \cos \theta_{f_i, f_j}.$$

This matrix of cosines was numerically computed at the fitted fibroglandular parameters to be

```
1, 0.664, 0.453, 0.453, 0.664, 0.453, 0.664, 0.466, 0.466, 0.466, 0.655, 0.654, 0.566
0.664, 1, 0.320, 0.320, 0.471, 0.121, 0.471, 0.512, 0.512, 0.512, 0.683, 0.445, 0.557
0.453, 0.320, 1, 0.176, 0.320, 0.176, 0.121, 0.131, 0.131, 0.131, 0.175, 0.114, 0.500
0.453, 0.320, 0.176, 1, 0.121, 0.176, 0.320, 0.131, 0.131, 0.131, 0.202, 0.322, 0.500
0.664, 0.471, 0.320, 0.121, 1, 0.320, 0.471, 0.512, 0.512, 0.512, 0.683, 0.445, 0.557
0.453, 0.121, 0.176, 0.176, 0.320, 1, 0.320, 0.131, 0.131, 0.131, 0.202, 0.322, 0.500
0.664, 0.471, 0.121, 0.320, 0.471, 0.320, 1, 0.512, 0.512, 0.512, 0.756, 0.996, 0.557
0.466, 0.512, 0.131, 0.131, 0.512, 0.131, 0.512, 1, 0.556, 0.556, 0.737, 0.481, 0.446
0.466, 0.512, 0.131, 0.131, 0.512, 0.131, 0.512, 0.556, 1, 0.556, 0.737, 0.481, 0.446
0.655, 0.683, 0.175, 0.202, 0.683, 0.202, 0.756, 0.737, 0.737, 1, 0.723, 0.608
0.654, 0.445, 0.114, 0.322, 0.445, 0.322, 0.996, 0.481, 0.481, 0.481, 0.723, 1, 0.541
0.566, 0.557, 0.500, 0.500, 0.557, 0.500, 0.557, 0.446, 0.446, 0.446, 0.608, 0.541, 1
```

The fact that none of the off-diagonal values are 1 or -1 indicates that our sensitivities are functionally independent pairwise, but we must consider their independence as a group to determine if they are linearly independent.

To establish the independence of the sensitivity functions on

$\mathcal{D}$

, we can compute the volume of the hyper-parallelepiped spanned by the normalized functions. This value will be 0 if the functions are linearly dependent, and will be 1 if the functions are mutually orthogonal, but interpreting intermediate values is more difficult. Due to the relationship between hyper-volumes and dimension, for a large number of functions, this value will rapidly become small if these functions are not perfectly orthogonal.

Therefore, to eliminate the effect of dimensional scaling on this measure, we take the n -th

root of this volume to get the side length of a hyper-cube with the same volume as this parallelepiped, and we use this effective length ratio as the relevant measure of independence.

The volume of the hyper-parallelepiped can be computed as the square root of the determinant of the matrix of cosines. The volume of the unit hyper-parallelepiped with these directional cosines is the same as that of a hyper-cube with side length 0.592, indicating that these functions, while not perfectly independent, are sufficiently independent to justify their use. Removing the rows and columns corresponding to the parameters

a

and

b

, we obtain an equivalent hyper-cube length of 0.752, indicating that the underlying sensitivities for adipose tissue are more independent than those for the fibroglandular tissue. Alternatively, one could consider a unit-parallelepiped spanning the same volume, but with uniform angles between its edges. This equivalent parallelepiped gives an effective “average angle” between the sensitivities. Doing this yields an effective angle of 42.5° between the fibroglandular sensitivities, and 48.6° between the adipose sensitivities.

By either of these metrics, we see that at the fitted values we obtained, and over the strain range we have considered, our model depends on our parameters independently. Therefore, while there is some covariance between our parameters, this covariance is not so severe as to justify a reparameterization of our model, or a reduction of the number of parameters.

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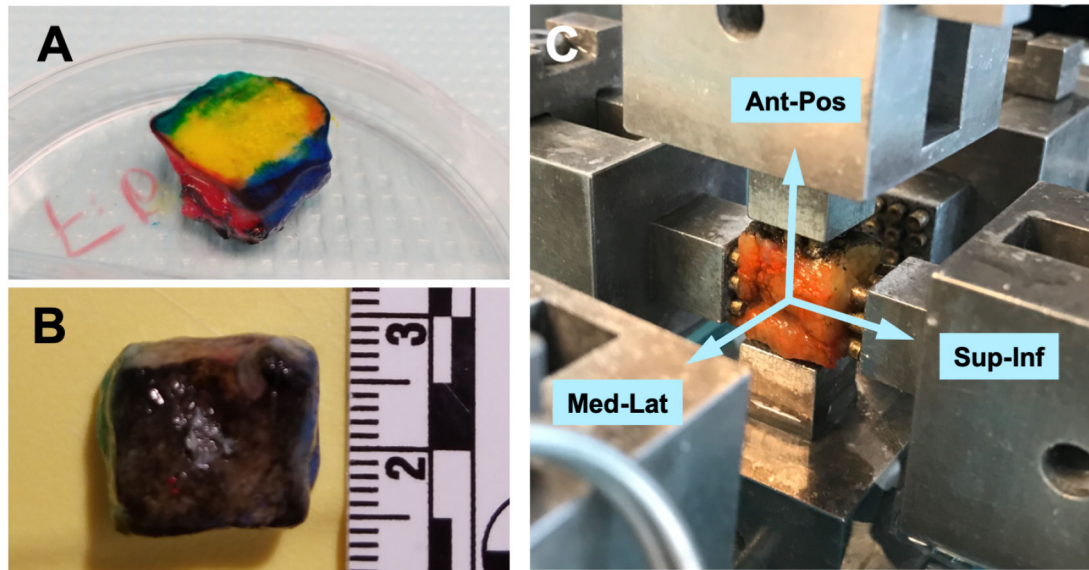


Fig. 1:

(A) An example of the a human breast tissue specimen with inert dyes applied to various sides to allow anatomical orientation identification. (B) Final trimmed test specimen. (C) Test specimen placed in the triaxial testing device showing the anatomical axes.

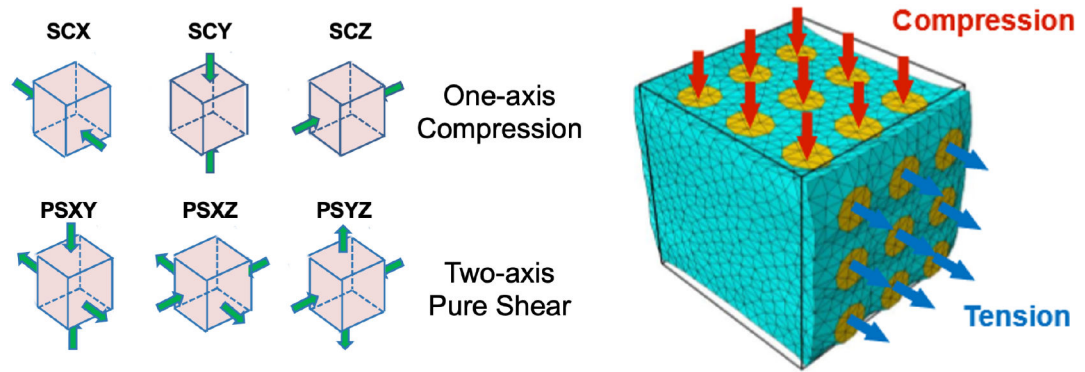
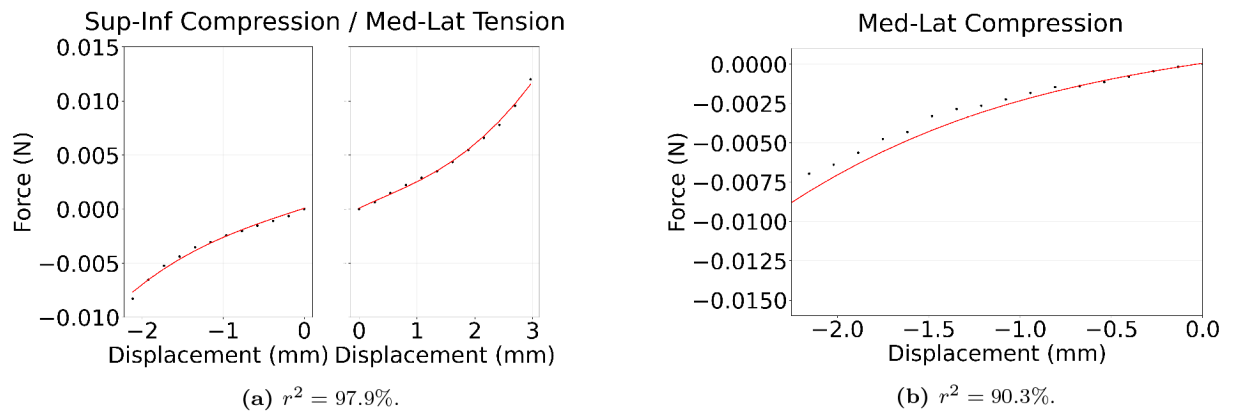
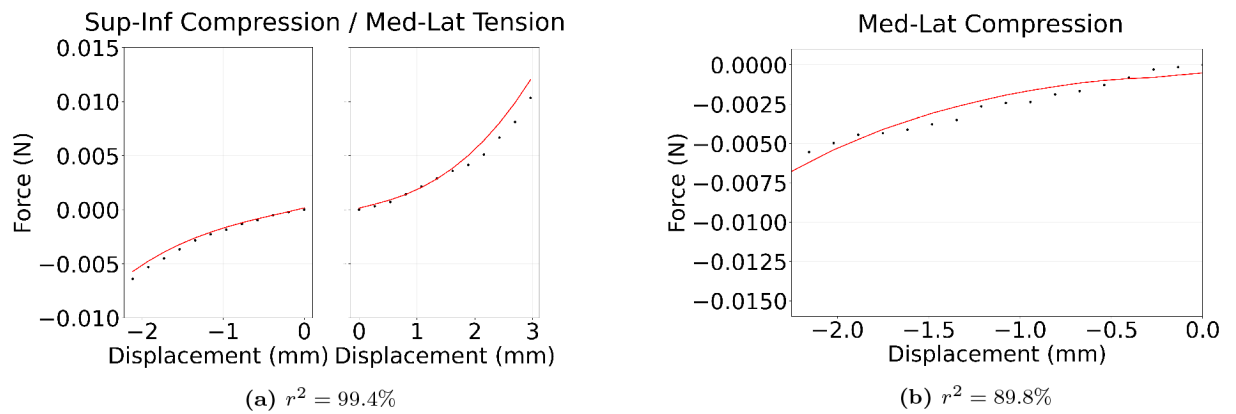


Fig. 2: Schematic of the applied 3D deformation configurations, where SC=simple compression and PS=pure shear. Also shown is representative finite element model of the simulated specimen loading configuration.

**Fig. 3:**

The model was able to accurately fit the experimental adipose data used to determine material constants (a), and predict experimental results that were not included in fitting data (b).

**Fig. 4:**

The model with exponential augmentation was able to accurately fit the experimental fibroglandular data used to determine material constants (a), and predict experimental results that were not included in fitting data (b).

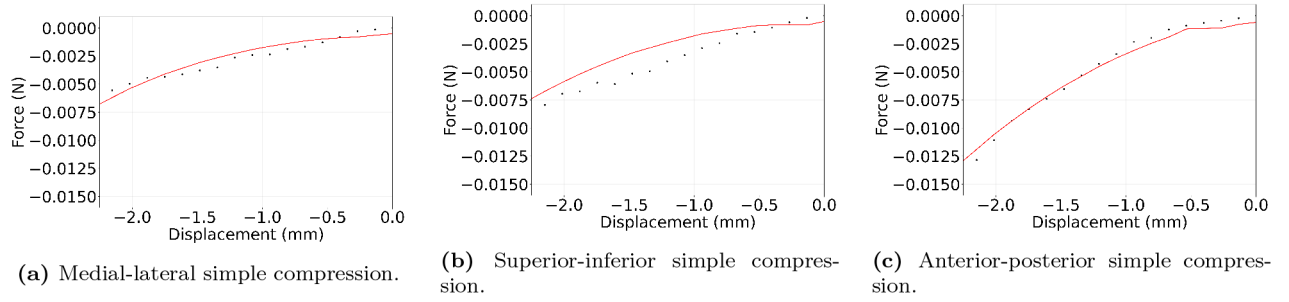


Fig. 5:
Fibroglandular simple compression data (black) against fitted model predictions (red) demonstrates greater stiffness in the anterior-posterior direction.

Table 1:

Deformation protocols used in this study together with the corresponding homogeneous deformation gradient.

Parameters

λ

and

η

are the principal stretches greater than 1, where incompressibility has been imposed to reduce the number of free principal stretches. Simple compression modes have traction free boundary conditions applied to uncompressed faces, meaning that

$(\lambda\eta)^{-1}$

is imposed while

λ

and

η

are determined by the zero traction boundary conditions. For pure shear modes, the stretches

λ

and

λ^{-1}

are imposed, with the remaining principal stretch determined to be 1 by incompressibility.

Protocol	Deformation	F_A^a
1	Med-Lat Tension / Sup-Inf Compression	$\text{Diag}(\lambda, \lambda^{-1}, 1)$
2	Med-Lat Tension / Ant-Pos Compression	$\text{Diag}(\lambda, 1, \lambda^{-1})$
3	Sup-Inf Tension / Ant-Pos Compression	$\text{Diag}(1, \lambda, \lambda^{-1})$
4	Med-Lat Simple Compression	$\text{Diag}((\lambda\eta)^{-1}, \lambda, \eta)$
5	Sup-Inf Simple Compression	$\text{Diag}(\lambda, (\lambda\eta)^{-1}, \eta)$
6	Ant-Pos Simple Compression	$\text{Diag}(\lambda, \eta, (\lambda\eta)^{-1})$

Table 2:

Fitted model parameters for adipose and fibroglandular tissue.

Parameter	Adipose	Fibroglandular
μ	49.4	65.5
M_{11}	193	191
M_{22}	253	348
M_{33}	226	206
M_{12}	207	351
M_{13}	195	194
M_{23}	140	102
M_{44}	145	158
M_{55}	273	242
M_{66}	138	116
a	0	1.59
b	0	6.13
K	5730	5520